Lean Production

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In this chapter, we examine several additional approaches related to methods engineering. These approaches are aimed at increasing productivity, reducing cost, improving quality, and solving problems in the operations of an organization. These objectives are basically the same as those in methods engineering. In many ways, the topics discussed here are adaptations and modifications of the previous methods engineering techniques and principles. However, these alternative approaches are of more recent vintage than the traditional methods engineering programs, and they represent a more contemporary view of work and the social environment in which it is performed. One of the distinguishing characteristics of this current view is that the active participation of the worker should be enlisted in the improvement activities. The new approaches can be presented in two ways: (1) lean production, a highly efficient production system developed at Toyota Motors in Japan, and (2) Six Sigma, a quality improvement procedure that seeks perfection.

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Lean production means doing more work with fewer resources. It is an adaptation of mass production in which work is accomplished in less time, smaller space, with fewer workers and less equipment, and yet achieving higher quality levels in the final product. Lean production also means giving customers what they want and satisfying or surpassing their expectations. Lean production is based on the way manufacturing operations were organized at the Toyota Motors Company in Japan during the 1980s. Toyota had pioneered a system of production that was quite different from the mass production techniques used by automobile companies in the United States and Europe. The Toyota Production System, as it was called before the term "lean" was applied to it, began to evolve in the 1950s to cope with the realities of Japan's postwar economy (e.g., a much smaller domestic auto market than in the United States or Europe and a scarcity of Japanese investment capital for plant and equipment). To deal with these challenges, Toyota developed a production system that could produce a variety of car models with fewer quality problems, lower inventory levels, smaller manufacturing lot sizes for the parts used in the cars, and reduced lead times to produce the cars. Development of the Toyota Production System was led by Taiichi Ohno (Historical Note 1). In this chapter, we examine the methods used at Toyota that have come to be called lean production.

HISTORICAL NOTE 1 LEAN PRODUCTION

The person credited with developing the Toyota Production System is Taiichi Ohno (1912–1990), an engineer and executive at Toyota Motors. In the post–World War II period, the Japanese automotive industry had to basically start over. Ohno visited a U.S. auto plant to learn American production methods. However, the car market in Japan was much smaller than its U.S. counterpart, so a Japanese plant could not afford the large production runs and huge work-in-process inventories that were common in the United States. Ohno knew that Toyota's plants needed to be more flexible. Also, space was (and is) very precious in Japan. These conditions, as well as Ohno's aversion to waste in any form (***muda***, as the Japanese call it), motivated him to develop some of the basic ideas and procedures that have come to be known as lean production. He and his colleagues then proceeded to perfect these ideas and procedures over the next several decades, including just-in-time production and the ***kanban*** system of production control, production leveling, setup time reduction, quality circles, and dedicated adherence to quality control.

Ohno himself did not coin the term "lean production" to describe the collection of methods and systems used at Toyota to improve production efficiency. In fact, he titled his book ***The Toyota Production System: Beyond Large-Scale Production*** [13]. The term "lean production" was coined by researchers at Massachusetts Institute of Technology to describe the activities and programs that seemed to explain Toyota's success in the efficiency with which they produced cars and the quality of the cars they produced. In an MIT research project that came to be known as the International Motor Vehicle Program (IMVP), a survey was conducted of 87 automobile assembly plants throughout the world. The research was popularized by the book ***The Machine that Changed the World*** [23]. In the subtitle of the book was the term "lean production."

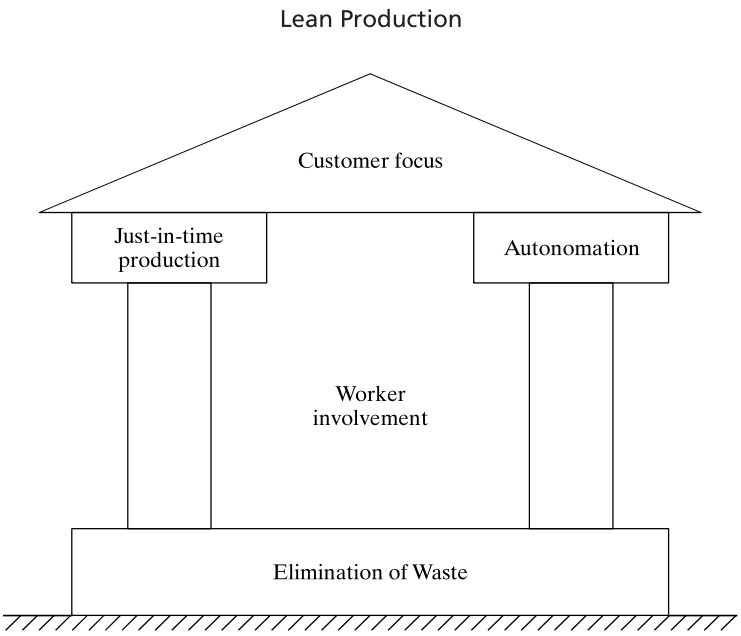

Figure 1 The structure of a lean production system.

TABLE 1 The Elements of Just-in-Time Production, Worker Involvement, and Autonomation in the Lean Production Structure

Just-in-Time Production	Worker Involvement	Autonomation
Pull system of production control using kanbans	Continuous improvement (kaizen)	Stop the process when something goes wrong (e.g., production of defects)
Setup time reduction for smaller batch sizes	Quality circles	Prevention of overproduction
Production leveling	Visual management	Error prevention and mistake proofing
On-time deliveries	The 5S system	Total productive maintenance for reliable equipment
Zero defects	Standardized work procedures	
Flexible workers	Participation in total productive maintenance by workers	

The ingredients of a lean production system are shown in the structure in Figure 1.[1] At the base of the structure is the foundation of the Toyota system: elimination of waste in production operations. Taiichi Ohno was driven to reduce waste at Toyota. Standing on the foundation are two pillars: (1) just-in-time production and (2) autonomation (automation with a human touch).[2] The two pillars support a roof that symbolizes a focus on the customer. The goal of lean production is customer satisfaction. Between the two

[1]Various forms of the "lean structure" have appeared in the literature. We believe the one shown here is representative.

[2]Taiichi Ohno describes just-in-time and autonomation as the two pillars needed to support the Toyota production system [13].

pillars and residing inside the structure is an emphasis on worker involvement: workers who are motivated, flexible, and continually striving to make improvements. Table 1 identifies the elements of just-in-time production, worker involvement, and autonomation in the lean production structure.

1 ELIMINATION OF WASTE

The underlying basis of the Toyota Production System is elimination of waste—called ***muda*** in Japanese. The very word has the sound of something messy (perhaps because it begins with the English word "mud"). In manufacturing, waste abounds. Ohno identified seven forms of waste in manufacturing that he wanted to eliminate by means of the various procedures that made up the Toyota system. Ohno's seven forms of waste are as follows:

1. Production of defective parts
2. Production of more than the number of items needed (overproduction)
3. Excessive inventories
4. Unnecessary processing steps
5. Unnecessary movement of people
6. Unnecessary transport and handling of materials
7. Workers waiting.

1.1 Production of Defective Parts

Eliminating production of defective parts (waste form 1) requires a quality control system that achieves perfect first-time quality. In the area of quality control, the Toyota production system was in sharp contrast with the traditional QC systems used in mass production. In mass production, quality control is typically defined in terms of an acceptable quality level (AQL), which means that a certain minimum level of fraction defects is sufficient, even satisfactory. In lean production, by contrast, perfect quality is required. The just-in-time delivery discipline used in lean production necessitates a zero defects level in parts quality, because if the part delivered to the downstream workstation is defective, production is forced to stop. There is little or no inventory in a lean system to act as a buffer. In mass production, inventory buffers are used just in case these quality problems occur. The defective work units are simply taken off the line and replaced with acceptable units. However, the problem is that such a policy tends to perpetuate the cause of the poor quality. Therefore, defective parts continue to be produced. In lean production, a single defect draws attention to the quality problem, forcing corrective action and a permanent solution. Workers inspect their own production, minimizing the delivery of defects to the downstream production station.

1.2 Overproduction and Excessive Inventories

Overproduction (waste form 2) and excessive inventories (waste form 3) are correlated. Producing more parts than necessary means that there are leftover parts that must be stored. Of all of the forms of muda, Ohno believed that the "greatest waste of all is

excess inventory."[3] Overproduction and excess inventories generate increased costs in the following areas:

- Warehousing (building, lighting and heating, maintenance)
- Storage equipment (pallets, rack systems, forklifts)
- Additional workers to maintain and manage the extra inventory
- Additional workers to make the parts that were overproduced
- Other production costs (raw materials, machinery, power, maintenance) to make the parts that were overproduced and stored
- Interest payments to finance all of the above.

The kanban system (Section 2.1) for just-in-time production provides a control mechanism at each workstation to produce only the minimum quantity of parts needed to feed the next process in the sequence. In so doing, it limits the amount of inventory that is allowed to accumulate between operations.

1.3 Other Forms of Waste

Waste forms 4 through 7 all represent inefficient use of resources. Most of these inefficiencies can be corrected through the use of good methods engineering and layout planning. Good equipment maintenance (total productive maintenance, Section 3.3) helps to reduce machine downtime that causes waiting time for workers.

Unnecessary Processing Steps. Unnecessary processing steps mean that energy is being expended by the worker and/or machine to accomplish work that adds no value to the product. An example of this waste form is a product that is designed with features that serve no useful function to the customer, and yet time and cost are consumed to create those features. Another reason for unnecessary processing steps is that the processing method for the given task has not been well designed. Perhaps no work design has occurred at all. Consequently, the method used for the task includes wasted hand and body motions, unnecessary work elements, inappropriate hand tools, inefficient production equipment, poor ergonomics, and safety hazards.

Unnecessary Movement of Workers and Materials. The movement of people and materials is a necessary activity in manufacturing. Body motions and walking are necessary and natural elements of the work cycle for most workers, and materials must be transported from operation to operation during their processing. It is when the movement of workers or materials is done unnecessarily and without adding value to the

[3]Ohno [13].

product that waste occurs. The following reasons explain why people and materials are sometimes moved unnecessarily:

- *Inefficient workplace layout.* Tools and parts are randomly organized in the work space, so that workers must search for what they need and use inefficient motion patterns to complete their tasks.
- *Inefficient plant layout.* Workstations are not arranged along the line of flow of the processing sequence.
- *Improper material-handling method.* For example, manual-handling methods are used instead of mechanized or automated equipment.
- *Improper spacing of production machines.* Greater distances mean longer transit times between machines.
- *Larger equipment than necessary for the task.* Larger machines need larger access space and greater distances between machines.
- *Conventional batch production.* In batch production, changeovers are required between batches that result in downtime during which nothing is produced.

Workers Waiting. The seventh form of muda is workers waiting. When workers are forced to wait, it means that no work (either value-adding or non–value-adding) is being performed. There is a variety of reasons why workers are sometimes forced to wait:

- Materials have not been delivered to the workstation.
- The assembly line has stopped.
- A machine has to be repaired.
- A machine is being serviced by the setup crew.
- A machine is performing its automatic processing cycle on a work part.

2 JUST-IN-TIME PRODUCTION

Just-in-time (JIT) production systems were developed to minimize inventories, especially work-in-process. Excessive WIP is seen in the Toyota Production System as waste that should be minimized or eliminated. The ideal ***just-in-time production system*** produces and delivers exactly the required number of each component to the downstream operation in the manufacturing sequence at the exact moment when that component is needed. Each component is delivered "just in time." This delivery discipline minimizes WIP and manufacturing lead time as well as the space and money invested in WIP. In the Toyota Production System, the just-in-time discipline was applied not only to its own production operations but to supplier delivery operations as well.

While the development of JIT production systems is attributed to Toyota, many U.S. firms have also adopted the just-in-time philosophy. Other terms are sometimes applied to the American practice of JIT to suggest differences with the Japanese practice. For example, ***continuous flow manufacturing*** is a widely used term in the United States that denotes a just-in-time style of production operations in which work parts are processed and transported directly to the next workstation one unit at a time. Each process

is completed just before the next process in the sequence begins. In effect, this is JIT with a batch size of one work unit. Prior to JIT, the traditional U.S. practice might be described as a "just-in-case" philosophy—that is, to hold large in-process inventories to cope with production problems such as late deliveries of components, machine breakdowns, defective components, and wildcat strikes.

The just-in-time production discipline has shown itself to be very effective in high-volume repetitive operations, such as those found in the automotive industry [12]. The potential for WIP accumulation in this type of manufacturing is significant, due to the large quantities of products made and the large numbers of components per product.

The principal objective of JIT is to reduce inventories. However, inventory reduction cannot simply be mandated to happen. Three requisites must be in place for a just-in-time production system to operate successfully: (1) a pull system of production control, (2) setup time reduction for smaller batch sizes, and (3) stable and reliable production operations.

2.1 Pull System of Production Control

JIT is based on a ***pull system*** of production control, in which the order to make and deliver parts at each workstation in the production sequence comes from the downstream station that uses those parts. When the supply of parts at a given workstation is about to be exhausted, that station orders the upstream station to replenish the supply. Only upon receipt of this order is the upstream station authorized to produce the needed parts. When this procedure is repeated at each workstation throughout the plant, it has the effect of pulling parts through the production system. By comparison, in a ***push system*** of production control, parts at each workstation are produced irrespective of the immediate need for those parts at its respective downstream station. In effect, this production discipline pushes parts through the plant. The risk in a push system is that more parts get produced in the factory than it can handle, resulting in large queues of work in front of machines. The machines are unable to keep up with arriving work, and the factory becomes overloaded with work-in-process inventory. The difference between push and pull systems of production control is illustrated in Figure 2.

The Toyota Production System implemented its pull system by means of kanbans. The word *kanban* (pronounced kahn-bahn) means "card" in Japanese. The ***Kanban system*** of production control is based on the use of cards that authorize (1) parts production and (2) parts delivery in the plant. Thus, there are two types of kanbans: (1) production kanbans, and (2) transport kanbans. A ***production kanban*** (P-kanban) authorizes the upstream station to produce a batch of parts. As parts are produced, they are placed in containers, so the batch quantity is just sufficient to fill the container. Production of more than this quantity of parts is not allowed in the kanban system. A ***transport kanban*** (T-kanban) authorizes transport of the container of parts to the downstream station.

The operation of a kanban system is illustrated in Figure 3. The workstations shown in the figure (station i and station $i + 1$) are only two in a sequence of multiple stations upstream and downstream. The flow of work is from station i (the upstream station)

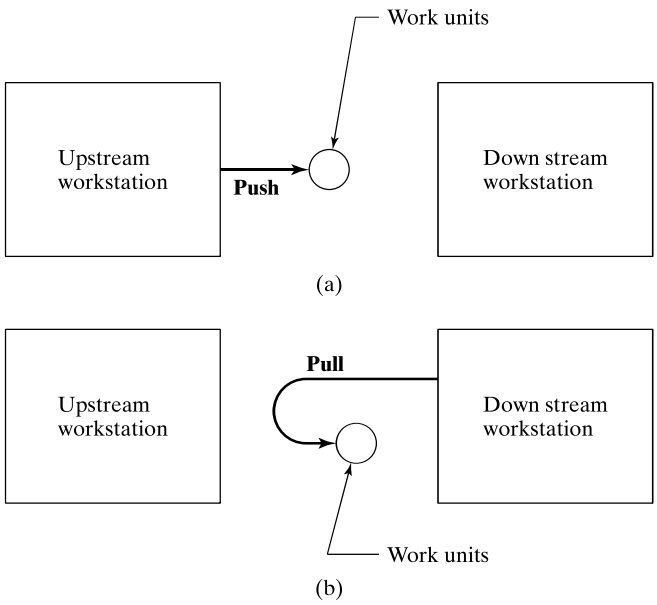

Figure 2 The difference between push and pull systems of production control: (a) push system, in which upstream stations push work to downstream stations and (b) pull system, in which downstream stations pull work from upstream stations.

to station $i + 1$ (the downstream station). The sequence of steps in the kanban pull system is as follows (our numbering sequence is coordinated with Figure 3):

1. Station $i + 1$ removes the next P-kanban from the dispatching rack. This P-kanban authorizes it to process a container of part b. A material handling worker removes the T-kanban from the incoming container of part b and takes it back to station i.
2. At station i, the material-handling worker finds the container of part b, removes the P-kanban and replaces it with the T-kanban. He then puts the P-kanban in the dispatching rack at station i.
3. The container of part b that was at station i is moved to station $i + 1$, as authorized by the T-kanban. The P-kanban for part b at station i authorizes station i to process a new container of part b, but it must wait its turn in the rack for the other P-kanbans ahead of it. Scheduling of work at each station is determined by the order in which the production kanbans are placed in the dispatching rack. Meanwhile, processing of the b parts at station $i + 1$ has been completed and that station removes the next P-kanban from the dispatching rack and begins processing that container of parts (it happens to be part d as indicated in the figure).

As mentioned, stations i and $i + 1$ are only two adjacent stations in a longer sequence. All other pairs of upstream and downstream stations operate according to the

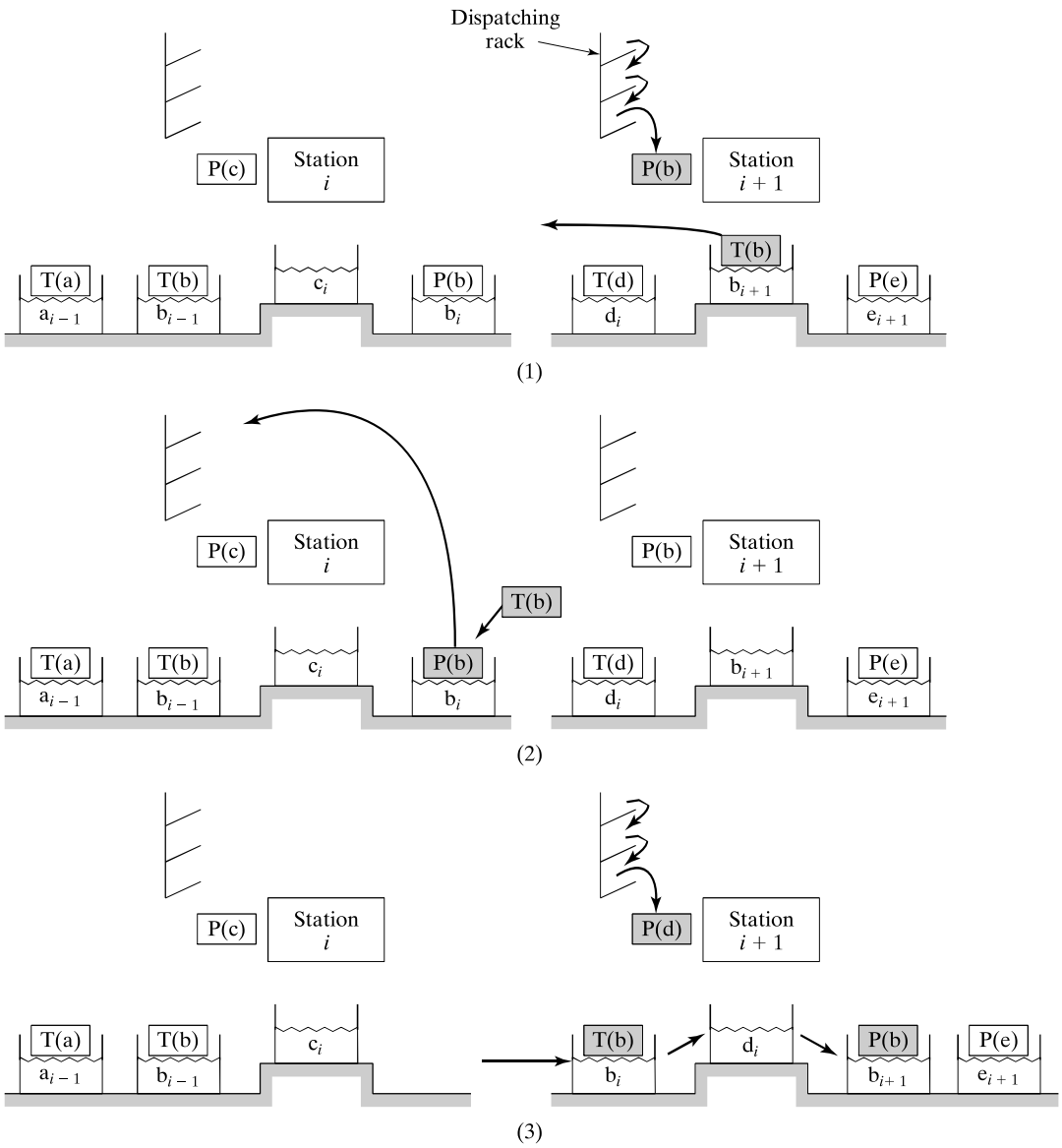

Figure 3 Operation of a kanban system between workstations (see description of steps in the text).

same kanban pull system. This production control system avoids unnecessary paperwork. The kanban cards are used over and over again instead of generating new production and transport orders every cycle. Although considerable labor is involved in material handling (moving cards and containers between stations), this is claimed to promote teamwork and cooperation among workers.

Today's kanban installations in the automotive industry rely on modern communications technologies rather than cards [6]. These electronic kanban systems connect

production workers to the material handlers who deliver the parts. For example, one system installed at Ford Motor Company uses battery-powered wireless buttons located at each operator workstation. When the supply of parts gets down to a certain level, the operator presses the button, which signals the material handlers to deliver another batch of parts. Each button emits a low power signal that is received by antennas attached to the plant ceiling and transmitted to a computer system that provides instructions to the material handlers about what to deliver, where, and when.

2.2 Setup Time Reduction for Smaller Batch Sizes

To minimize work-in-process inventories in manufacturing, batch sizes and setup times must be minimized. The relationship between batch size and setup time is given by the economic order quantity formula. In our mathematical model for total inventory cost, from which the EOQ formula is derived, average inventory level is equal to one-half the batch size. To reduce average inventory level, batch size must be reduced. And to reduce batch size, setup cost must be reduced. This means reducing setup times. Reduced setup times permit smaller batches and lower work-in-process levels.

Setup time reductions result from a number of basic approaches that are best described as methods improvements. The approaches are largely credited to the pioneering work of Shigeo Shingo, an industrial engineering consultant at Toyota Motors during the 1960s and 1970s.[4]

The starting point in setup time reduction is to recognize that the work elements in setting up a machine can be separated into two categories:

- *Internal elements.* These work elements can only be done while the production machine is stopped.
- *External elements.* These work elements do not require that the machine be stopped.

Examples of the two categories are listed in Table 2. By their nature, external work elements can be accomplished while the previous job is still running. For the setup time to be reduced in a given changeover, the setup tooling (e.g., die, fixture, mold) must be designed and the changeover procedure must be planned to permit as much of the setup as possible to consist of external work elements.

Although it is desirable to reduce the times required to accomplish both internal and external work elements, the internal elements must be given a higher priority, since they determine the length of time that a machine will not be producing during a changeover. The following approaches apply mostly to the internal elements in a setup:

- Use time and motion study and methods improvements to reduce the sum of the internal work elements to the minimum possible time.
- Use two workers working in parallel to accomplish the setup, rather than one worker working alone. This approach is not applicable to all changeover situations,

[4]They have been documented in [4], [12], [18], [19], and [22]. The approaches can be applied to virtually all batch production situations, but their applications in the automotive industry have emphasized pressworking and machining operations, owing to their widespread use in this industry.

TABLE 2 Examples of Internal and External Work Elements During a Production Setup or Changeover

Internal Work Elements	External Work Elements
Removing tooling (e.g., dies, molds, fixtures) used in previous production job from machine	Retrieving tooling for next job from tool storage
Positioning and attaching tooling for next job in machine	Assembling tooling components next to machine (if tooling consists of separate pieces that must be assembled)
Making final adjustments and alignments of tooling	Reading engineering drawings regarding new setup
Trying out the setup and making trial parts	Reprogramming machine for next job (e.g., downloading part program for new part)

TABLE 3 Examples of Setup Reductions in Japanese and U.S. Industries

Industry	Equipment Type	Setup Time Before Reduction	Setup Time After Reduction	Reduction (%)
Japanese automotive	1000 ton press	4 hr	3 min	98.7
Japanese diesel	Transfer line	9.3 hr	9 min	98.4
U.S. power tool	Punch press	2 hr	3 min	97.5
Japanese automotive	Machine tool	6 hr	10 min	97.2
U.S. electric appliance	45 ton press	50 min	2 min	96.0

Source: [21].

but where it is applicable, it theoretically reduces the downtime to one-half the time required for one worker.

- Eliminate or minimize adjustments in the setup. Adjustments are time-consuming.
- Use quick-acting fasteners instead of bolts and nuts where possible.
- When bolts and nuts must be used with washers, use U-shaped instead of O-shaped washers. A U-shaped washer can be inserted between the parts to be clamped without completely disassembling the nut from the bolt. To add an O-shaped washer, the nut must be removed from the bolt.
- Design modular fixtures consisting of a base unit plus insert tooling that can be quickly changed for each new part style. The base unit remains attached to the production machine, so that only the insert tooling must be changed.

Although methods for reducing setup time were pioneered by the Japanese, U.S. firms have also adopted these methods. Results of the efforts are sometimes dramatic. Table 3 presents some examples of setup reductions in Japanese and U.S. industries reported by Suzaki [21].

The economic impact of setup time reduction in a production operation can be assessed using the economic order quantity equations.

Example 1 Effect of Setup Reduction on EOQ and Inventory Cost

Let us determine the effect on economic batch size and total inventory costs of reducing setup time given the following information: annual demand = 15,000 pc/yr, unit cost = $20, holding cost rate = 18%/yr, setup time = 5 hr, and cost of downtime during setup = $150/hr. Suppose it were possible to reduce setup time from 5 hr to 5 min (this kind of reduction is not so far-fetched, given the data in Table 3). Determine (a) the economic order quantity and (b) total inventory cost for this new situation.

Solution: First, recall the results of the earlier example. Setup cost C_{su} = (5 hr)($150/hr) = $750 and holding cost C_h = 0.18($20) = $3.60. Using these values, the economic order quantity was computed as

$$EOQ = \sqrt{\frac{2(15{,}000)(750)}{3.60}} = 2500 \text{ units}$$

The corresponding total inventory costs are computed as

$$TIC = 0.5(2500)(3.60) + 750(15{,}000/2500) = \$9000$$

(a) Reducing the setup time to 5 min reduces the setup cost to C_{su} = (5/60 hr)($150) = $12.50. Holding cost remains the same, and the new economic order quantity is

$$EOQ = \sqrt{\frac{2(15{,}000)(12.50)}{3.60}} = 323 \text{ units}$$

This is a significant reduction from the 2500-piece batch size when setup time was 5 hr.

(b) Total inventory costs are also reduced, as follows:

$$TIC = 0.5(323)(3.60) + 12.50(15{,}000/323) = \$1161$$

This is an 87% cost reduction from the previous value. ■

2.3 Stable and Reliable Production Operations

Other requirements for a successful JIT production system include (1) production leveling, (2) on-time delivery, (3) defect-free components and materials, (4) reliable production equipment, (5) a workforce that is capable, committed, and cooperative, and (6) a dependable supplier base.

Production Leveling. If production is to flow as smoothly as possible, there must be minimum perturbations from the fixed schedule. Perturbations in downstream operations tend to be magnified in upstream operations. A 10% change in final assembly is often amplified into a 50% change in parts production operations, due to overtime, unscheduled setups, variations from normal work procedures, and other exceptions. By maintaining a constant master production schedule over time, smooth workflow is achieved and disturbances in production are minimized.

The trouble is that demand for the final product is not constant. Accordingly, the production system must adjust to the ups and downs of the marketplace using ***production***

leveling, which means distributing the changes in product mix and quantity as evenly as possible over time. Production leveling can be accomplished using the following approaches [5]:

- Authorizing overtime during busy periods
- Using finished product inventories to absorb daily ups and downs in demand
- Adjusting the cycle times of the production operations
- Producing in small batch sizes that are enabled by setup time reduction techniques. In the ideal, the batch size is reduced to one. Instead of producing parts A and B according to a schedule that looks like this:

  AAAAAAAAAABBBBBBBBBB,

  the parts are instead scheduled like this:

  ABABABABABABABABABAB.

The benefits of production leveling include greater responsiveness to changes in product demand, shorter lead times, smaller in-process and finished goods inventories, and regularity in the workload of production workers.

On-Time Deliveries, Zero Defects, and Reliable Production Equipment. Just-in-time production requires near perfection in on-time delivery, parts quality, and equipment reliability. Owing to the small lot sizes used in JIT, parts must be delivered before stockouts occur at downstream stations. Otherwise, these stations are starved for work and production is forced to stop.

JIT requires high quality in every aspect of production. If parts are produced with quality defects, they cannot be used in subsequent processing or assembly stations, thus interrupting work at those stations and possibly stopping production. Such a severe penalty motivates a discipline of very high quality levels (zero defects) in parts fabrication. Workers are trained to inspect their own output to make sure it is right before it goes to the next operation. In effect, this means controlling quality during production rather than relying on inspectors to discover the defects later.

JIT also requires highly reliable production equipment. Low work-in-process leaves little room for equipment stoppages. Machine breakdowns cannot be tolerated in a JIT production system. The equipment must be "designed for reliability," and the plant that operates the equipment must employ total productive maintenance.

Workforce and Supplier Base. Workers in a just-in-time production system must be cooperative, committed, and cross-trained. Small batch sizes mean that workers must be willing and able to perform a variety of tasks and to produce a variety of part styles at their workstations. As indicated above, they must be inspectors as well as production workers in order to ensure the quality of their own output. They must be able to deal with minor technical problems that may be experienced with the production equipment, so that major breakdowns are avoided.

The suppliers of raw materials and components to the company must be held to the same standards of on-time delivery, zero defects, and other JIT requirements as

the company itself. New policies such as the following are required for JIT for dealing with vendors:

- Reducing the total number of suppliers, thus allowing the remaining suppliers to do more business
- Entering into long-term agreements and partnerships with suppliers, so that suppliers do not have to worry about competitively bidding for every order
- Establishing quality and delivery standards and selecting suppliers on the basis of their capacity to meet these standards
- Placing employees into supplier plants to help those suppliers develop their own JIT systems
- Selecting parts suppliers that are located near the company's final assembly plant to reduce transportation and delivery problems.

3 AUTONOMATION

Appearing at first glance to be a misspelling of "automation" ***autonomation*** has been defined by Taiichi Ohno as "automation with a human touch" [13].[5] The notion is that the machines operate autonomously as long as they are functioning properly. When they do not function properly—for example, when they produce a defective part—they are designed to stop immediately. Another aspect of autonomation is that the machines and processes are designed to prevent errors, to be mistake-proof. Finally, machines in the Toyota Production System must be reliable, which requires an effective maintenance program. Three aspects of autonomation are discussed in this section: (1) stopping the process automatically when something is wrong, (2) error prevention, and (3) total productive maintenance.

3.1 Stopping the Process

Much of autonomation is embodied in the Japanese word ***jidoka***, which refers to machines that are designed to stop automatically when something goes wrong, such as a defective part being processed. Production machines in Toyota plants are equipped with automatic stop devices that activate when a defective work unit is produced.[6] Therefore, when a machine stops, it draws attention to the problem, requiring corrective action to be taken to avoid recurrences. Adjustments must be made to fix the machine, thereby eliminating or reducing subsequent defects and improving overall quality of the final product.

In addition to its quality control function, autonomation also refers to machines that are controlled to stop production when the required quantity (the batch size) has been completed, thus preventing overproduction (one of the seven forms of muda).

[5] An alternative definition of autonomation is "automation with a human mind" [12].

[6] The origins of ***jidoka*** in the Toyota Production System can be traced to Ohno's work experience early in his career in the textile industry, where the weaving machines were equipped with automatic stopping mechanisms that shut down the looms when abnormal operating conditions occurred. He implemented the idea at Toyota.

Although autonomation is often applied to automated production machines, it can also be used with manual operations as well. In either case, it consists of the following control devices: (1) a sensor to detect abnormal operation that would result in a quality defect, (2) a device to count the number of parts that have been produced, and (3) a means to stop the machine or production line when abnormal operation is detected or the required batch quantity has been completed.

The alternative to autonomation is when a production machine is not equipped with these control mechanisms and continues to operate abnormally, possibly completing an entire batch of defective parts before the quality problem is even noticed, or producing more parts than the quantity required at the downstream workstation. To avoid such a calamity in a plant that does not have automatic stop mechanisms on its machines, each machine must have a worker in continuous attendance to monitor its operation. Machines equipped with autonomation do not require a worker to be present all the time when it is functioning correctly. Only when it stops must the worker attend to it. This allows one worker to oversee the operation of multiple machines, thereby improving worker productivity significantly.

Because workers are called upon to service multiple machines, and the machines are frequently of different types, the workers must be willing and able to develop a greater variety of skills than those who are responsible for only a single machine type. The net effect of more versatile workers is that the plant becomes more flexible in its ability to shift workers around among machines and jobs to respond to changes in workload mix.

At Toyota, the jidoka concept is extended to its final assembly lines. Workers are empowered to stop the assembly line when a quality problem is discovered, using pull cords located at regular intervals along the line. Downtime on final assembly lines in the automotive industry is expensive. Managers desperately want to avoid it. They accomplish this by making sure that the quality problems that cause it are eliminated. Pressure is applied on the parts fabrication departments and suppliers to keep parts and subassemblies that are defective out of the final assembly area.

3.2 Error Prevention

This aspect of autonomation is derived from two Japanese words: ***poka***, which means error, and ***yoke***, which means prevention. Together, ***poka-yoke*** refers to the prevention of errors through the use of low cost devices that detect and/or prevent them. The poka-yoke concept was developed by Shigeo Shingo, who also did pioneering work in setup time reduction (Section 2.2). The use of poka-yoke devices relieves the worker of constantly monitoring the process for errors that might cause defective parts or other undesirable consequences (e.g., unsafe working conditions, omitting a processing step).

Mistakes in manufacturing are common, and they often result in the production of defects. Examples include the following:

- Omission of processing steps, such as neglecting to spray lubricant on a mold cavity to prevent sticking of the molded part
- Incorrectly locating a work part in a fixture
- Using the wrong tool (for example, using the wrong cutting tool material for a given work material)

- Not aligning jigs and fixtures properly on the machine tool table (which could result in the entire batch of parts being processed incorrectly)
- Neglecting to add a component part in an assembly

When a poka-yoke identifies that an error or other exception has occurred, it responds in either or both of the following ways:

- *Stops the process.* The mechanized or automated cycle is stopped when a problem is detected in the operation of a production machine. For example, a limit switch installed in a workholder detects that the workpiece is incorrectly located and is interlocked with the milling machine to prevent the process from starting.
- *Provides an alert.* This response provides an audible or visible warning signal that an error has occurred. This signal alerts the operator and perhaps other workers and supervisors about the problem. The use of andon boards (Section 4.2) is a means of implementing this type of response.

3.3 Total Productive Maintenance

Production equipment in the Toyota Production System must be highly reliable. The just-in-time delivery system cannot tolerate machine breakdowns, since there is little buffer stock between workstations to keep upstream and downstream stations producing when the middle station stops. Lean production requires an equipment maintenance program that minimizes machine breakdowns. ***Total productive maintenance*** (TPM) is a coordinated group of activities whose objective is to minimize production losses due to equipment failures, malfunctions, and low utilization through the participation of workers at all levels of the organization. Worker teams are formed to solve maintenance problems (such as the kaizen projects, discussed in Section 4.1). Workers who operate equipment are assigned the routine tasks of inspecting, cleaning, and lubricating their machines. This leaves the regular maintenance workers with time to perform the more demanding technical duties, such as emergency maintenance, preventive maintenance, and predictive maintenance (see definitions in Table 4). In TPM, the goal is zero breakdowns.

TABLE 4 Some Maintenance Definitions

Term	Definition
Emergency maintenance	Repairing equipment that has broken down and returning it to operating condition. Action must be taken immediately to correct the malfunction. Also known as reactive maintenance.
Preventive maintenance	Performing routine repairs on equipment (e.g., replacement of key components) to prevent and avoid breakdowns.
Predictive maintenance	Anticipating equipment malfunctions before they occur based on computerized machine monitoring, machine operator being attentive to the way the machine is running, historical data, and other predictive techniques.
Total productive maintenance	Integration of preventive and predictive maintenance to avoid emergency maintenance.

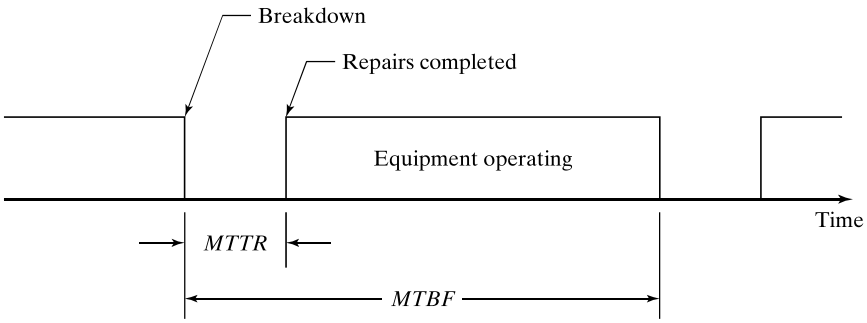

Figure 4 Time scale showing MTBF and MTTR for a piece of machine that breaks down occasionally.

The traditional measure of machine reliability is ***availability***, which is the proportion of the total desired operating time that the machine is actually available and operating.[7] To be more precise, availability is defined in terms of two other reliability measures: ***mean time between failures*** (*MTBF*) and ***mean time to repair*** (*MTTR*). *MTBF* indicates the average length of time the piece of equipment runs between breakdowns. *MTTR* indicates the average time required to service the equipment and put it back into operation after a breakdown occurs. *MTBF* and *MTTR* are illustrated in Figure 4. Using these terms, ***availability*** is defined as follows:

$$A = \frac{MTBF - MTTR}{MTBF} \tag{1}$$

where A = availability, $MTBF$ = mean time between failures, hr; and $MTTR$ = mean time to repair, hr. Availability is typically expressed as a percentage. When a piece of equipment is brand new (and being debugged), and later when it begins to age, its availability tends to be lower. This results in a typical U-shaped curve for availability as a function of time over the life of the equipment, as shown in Figure 5.

There are other reasons besides breakdowns that might cause a piece of production equipment to operate at less than its full capability. Such reasons include low equipment utilization, production of defective parts, and operating at less than the machine's designed speed.

Utilization refers to the amount of output of a production machine during a given time period (e.g., week) relative to its capacity during that same period. This can be expressed as an equation,

$$U = \frac{Q}{PC} \tag{2}$$

where U = utilization of the facility; Q = actual quantity produced by the facility during a given time period—for instance, pc/wk; and PC = production capacity for the same period, pc/wk. Utilization can also be measured as the number of hours of productive

[7]Availability is similar to line efficiency or uptime proportion.

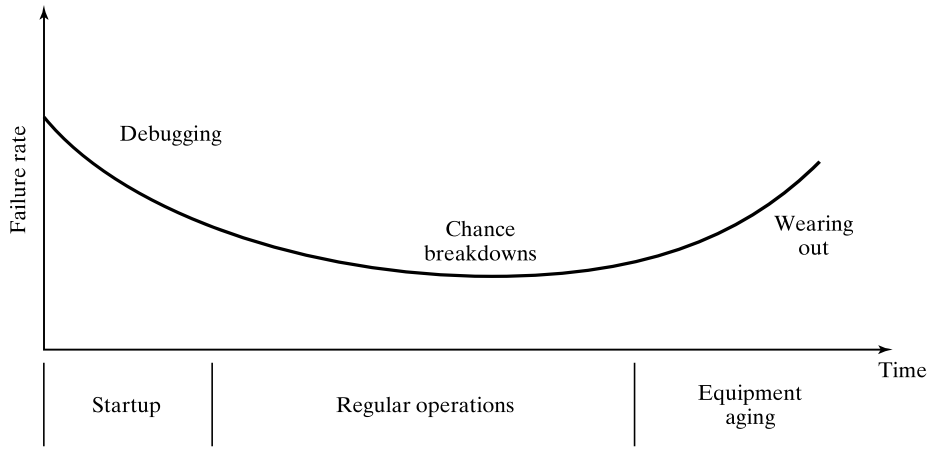

Figure 5 Typical U-shaped availability curve for a piece of equipment during its life.

operation relative to the total number of hours the machine is available. Reasons for poor machine utilization include poor scheduling of work, machine starved for work by upstream operation, downtime due to setups and changeovers between production batches, worker absenteeism, and low demand for the type of process performed by the machine. Utilization can be assessed for a single machine, an entire plant, or any other productive resource (e.g., labor). It is often expressed as a percentage (e.g., the plant is operating at 83% of capacity).

Production of defective parts may be due to incorrect machine settings, inaccurate adjustments in the setup, or improper tooling. All of these reasons are related to equipment maintenance problems. Additional reasons for producing defects that may not be related to equipment problems include defective starting materials and human error. The ***fraction defect rate*** is defined as the proportion of defective parts that are produced in a given process. The yield of an individual process is defined as

$$Y = 1 - q \tag{3}$$

where Y = process yield (ratio of conforming parts produced to total parts processed).

Finally, running the equipment at less than its designed speed also reduces its ***operating capability***, which is the ratio of the actual operating speed divided by the designed speed of the machine, symbolized as r_{os}.

All of these factors can be combined as follows to obtain a measure of the overall equipment effectiveness:

$$OEE = AUYr_{os} \tag{4}$$

where OEE = overall equipment effectiveness, A = availability, U = utilization, Y = process yield, and r_{os} = operating capability. The objective of total productive maintenance is to make OEE as close as possible to unity (100%).

4 WORKER INVOLVEMENT

Between the two pillars of lean production in Figure 1 are workers who are motivated and flexible and who participate in continuous improvement. Our discussion of worker involvement in the lean production system consists of three topics: (1) continuous improvement, (2) visual management and 5S, and (3) standardized work procedures.

4.1 Continuous Improvement

In the context of lean production, the Japanese word ***kaizen*** means continuous improvement of production operations. Kaizen is usually implemented by means of worker teams, sometimes called ***quality circles***,[8] which are organized to address specific problems that have been identified in the workplace. The teams address not only quality problems but also problems relating to productivity, cost, safety, maintenance, and other areas of interest to the organization. The term ***kaizen circle*** is also used, suggesting the broader range of issues that are usually involved in team activities.

Kaizen is a process that attempts to involve all workers as well as their supervisors and managers. Workers often are members of more than one kaizen circle. Although a principal purpose of organizing workers into teams is to solve problems in production, there are other less obvious but also important objectives. Kaizen circles encourage a an individual sense of responsibility, allow workers to gain acceptance and recognition among colleagues, and improve workers' technical skills [12].

Kaizen is applied on a problem-by-problem basis by worker teams. The team is constituted to deal with a specific problem. As mentioned above, the problem may relate to any of various areas of concern to the organization (e.g., quality, productivity, maintenance). Team members are selected according to their knowledge and expertise in the problem area and may be drawn from various departments. They serve part-time on a project team in addition to fulfilling their regular operational duties. On completion of the project, the team is disbanded. The usual expectation is that the team will meet two to four times per month, and each meeting will last about an hour.

The steps in each project depend on the type of problem being addressed. Details of the approaches recommended by different authors also vary [10, 15, 16]. Basically, the approach is the same as our systematic approach in methods engineering.

4.2 Visual Management and 5S

The principle behind ***visual management*** is that the status of the work situation should be evident just by looking at it.[9] If something is wrong, this condition should be obvious to the observer, so that corrective action can be taken immediately. Also called the ***visual workplace***, the principle applies to the entire plant environment. Objects that obstruct the view inside the plant are not allowed, making the entire interior space visible. The buildup of work-in-process is limited to a specified height (e.g., five feet). Thus, the visual workplace provides visibility throughout the plant and encourages continuous improvement and good housekeeping.

[8]The terms ***quality control circle*** and ***QC circle*** have also been used for these teams [9, 12].
[9]This is what statistician L. Tippett discovered in England during the 1920s while he was studying downtime of loom operations. The discovery resulted in the development of work sampling.

The use of kanbans in just-in-time production systems can be considered an example of visual management. The kanbans provide a visual mechanism for authorizing production and transportation of parts within the plant.

An important means of implementing visual management involves the use of an ***andon board***, a light panel positioned above a workstation or production line that is used to indicate its operating status. Its operation is commonly associated with the pull cords along a production line that permit a worker to stop the line. If a problem occurs, such as a line stoppage, the andon board identifies where the problem is and the nature of the problem. Different colored lights are often used to indicate the status of the operation. For example, a green light indicates normal operation, yellow means a worker has a problem and is calling for help, and red shows that the line has stopped. Other color codes may be used to indicate the end of a production run, shortage of materials, the need for a machine setup, and so on.

The visual workplace principle can also be applied in worker training. It includes the use of photographs, drawings, and diagrams to document work instructions, as opposed to lengthy text that is void of illustrations. "A picture is worth a thousand words" can be a powerful training tool for workers. In many cases, an actual example of the work part is used to convey the desired message; for example, examples of good parts and defective parts are shown to teach inspectors in quality control.

A means of involving workers in the visual workplace is a system called ***5S***, a set of procedures that is used to organize work areas in the plant. The name represents the first letters of five Japanese words as they would be spelled in English: ***seiri*** (sort), ***seiton*** (set in order), ***seiso*** (shine), ***seiketsu*** (standardize), and ***shitsuke*** (sustain).

The steps in 5S, briefly described in Table 5, provide an additional means of implementing visual management to provide a clean, orderly, and visible work environment that promotes high morale among workers and encourages continuous improvement. The steps are usually carried out by worker teams, and the 5S system must be a continuing process to sustain the accomplishments that have been made.

TABLE 5 The Five Steps in the 5S System

1. ***Sort.*** Sort things in the workplace; in particular, identify the items that are not used and dispose of them. This eliminates the clutter that usually accumulates in a workplace after many years.
2. ***Set in order.*** Organize the items remaining in the work area after sorting according to frequency of use, and provide easy access to those items that are most often needed.
3. ***Shine.*** Clean the work area and inspect it to make sure that everything is in its proper place. The alternative S-words for this step are "sweep" and "scrub," which may also be required in the work area.
4. ***Standardize.*** Document the standard locations for items in the workplace. For example, use a "shadow board" for hand tools, in which the outline of the tool is painted on the board to indicate where it belongs. Looking at the shadow board, workers can immediately tell whether a tool is present and where to return it.
5. ***Sustain.*** Establish a plan for sustaining the gains made in the previous four steps, and assign individual responsibilities to team members for maintaining a clean and orderly work environment. Make workers responsible for taking care of the equipment they operate, which includes cleaning and performing minor maintenance tasks.

Source: [5] and [14].

4.3 Standardized Work Procedures

Standardized work procedures are established in the Toyota Production System, using approaches that are similar to the traditional methods engineering techniques. And time study is used to determine the length of time that should be taken to complete a given work cycle. However, there are differences in the Toyota approach to the development of standard methods and standard times, and these differences are emphasized in the discussion that follows. The following objectives are part of the standardized work procedures at Toyota:

- Maintain high productivity—that is, accomplish the required amount of production using the fewest possible number of workers.
- Balance the workload among all processes.
- Minimize the amount of work-in-process in the production sequence.

In the Toyota Production System, a standardized work procedure for a given task has three components [13]: (1) cycle time, (2) work sequence, and (3) standard work-in-process quantity. These components are documented using forms that emphasize Toyota's unique manufacturing procedures. The forms are sometimes quite different from those used in traditional methods engineering and work measurement.

Cycle Time and Takt Time. The ***cycle time*** is the actual time it takes to complete a task. This time is established using stopwatch time study. Closely related to the cycle time is the ***takt time***, the reciprocal of the demand rate for a given product or part, adjusted for the available shift time in the factory ("takt" is a German word meaning cadence or pace). For a given product or part,

$$T_{takt} = \frac{EOT}{Q_{dd}} \tag{5}$$

where T_{takt} = takt time, EOT = effective daily operating time, and Q_{dd} = daily quantity of units demanded. The effective daily operating time is the shift hours worked each day, without subtracting any allowances for delays, breakdowns, or other sources of lost time. The daily quantity of units demanded is the monthly demand for the item divided by the number of working days in the month, without increasing the quantity to allow for defective units that might be produced. The reason why the effective daily operating time is not adjusted for lost time and the daily quantity is not adjusted for defects is to draw attention to these deficiencies so that corrective action will be taken to minimize or eliminate them.

Example 2 Takt Time

The monthly demand for a certain part is 10,000 units. There are 22 working days in the month. The plant operates two shifts, each with an effective operating time of 440 min. Determine the takt time for this part.

Solution: With two shifts, the effective daily operating time is 2(440 min) = 880 min. The daily quantity of units demanded is 10,000/22 = 454.5 pc/day.

$$T_{takt} = 880/454.5 = 1.94 \text{ min}$$

The takt time provides a specification based on demand for the part or product. In the Toyota Production System, the work must be designed so that the operation cycle time is synchronized with the takt time. This is accomplished through planning of the work sequence and standardizing the work-in-process quantity.

Work Sequence. The ***work sequence*** defines the order of work elements or operations performed by a given worker in accomplishing an assigned task. For a worker performing a repetitive work cycle at a single machine, it indicates the actions that need to be carried out, such as pick up the work part, load it into the machine, engage the feed, and unload the completed part at the end of the cycle. For a multifunction worker responsible for several machines, each with its own semiautomatic cycle, the work sequence describes what must be done at each machine and the order in which the machines must be attended. Pictures and drawings are often included (the visual workplace, Section 4.2) to show the proper use of hand tools and other aspects of the work routine such as safety practices and correct ergonomic posture.

For the multimachine situation, the work sequence is documented by means of the standard operations routine sheet, as shown in Figure 6. This form lists the machines that must be visited by the worker during each work cycle. A horizontal time scale indicates how long each operation should take, a solid line for the worker and dashed lines for the machines, and squiggly, nearly vertical lines are used to depict the worker walking between machines. The time lines show how well the machine cycle times are matched to the work routine performed by the worker each cycle.

Most production parts require more than one process (see sequential operations). It is not unusual for a dozen or more processing steps to be required to complete a part. The cycle time for a given process may be different from the takt time, which is based on the demand for the item produced in the task, not the requirements of the task itself. In the Toyota system, attempts are made to organize the work in such a way that the cycle time for each processing step is equal to the takt time for the part. By matching the cycle times with the takt time, all of the processes in the sequence are balanced, and the amount of work-in-process is minimized.

How can the work to produce a given part be organized so that the cycle time for each processing step is equal to the takt time? In the Toyota Production System, the

Sheet # 236		Date		Cycle time 1.8 min						Daily Quantity							Worker Name							
Op #	Operation	Time																						
	Machine #	Manual	Machine																					
1	M-23	0.25	1.31																					
2	M-16	0.25	1.44																					
3	M-68	0.25	0.87																					
4	D-47	0.20	0.36																					
5	D-33	0.20	0.54																					
6	D-25	0.20	0.62																					
7	D-42	0.20	0.67																					
	Time scale (min) →			0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	2.1

Figure 6 The standard operations routine sheet.

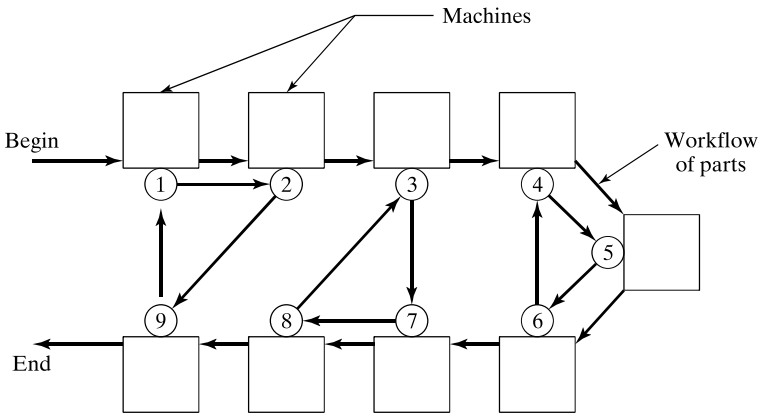

Figure 7 Cell layout with three workers performing nine operations. Operations are indicated by numbers in circles. The first worker performs operations 1, 2, and 9; the second worker performs operations 3, 7, and 8; and the third worker performs operations 4, 5, and 6.

tasks are integrated into work cells. At Toyota, a ***work cell*** is a group of workers and processing stations that are physically arranged in sequential order so that parts can be produced in small batches, in many cases one at a time. The cells are typically U-shaped rather than straight to promote comradeship among the workers and to achieve continuous flow of work in the cell. An example of a work cell layout and the operations performed in it by three workers is shown in Figure 7.[10] A total of nine operations are performed by the three workers, as depicted by the arrows in the figure. The corresponding standard operations routine sheets are illustrated in Figure 8. The total workload is balanced among the workers and the operations are allocated so the corresponding cycle times are equal to the takt time for the part produced in the cell.

Standard Work-in-Process Quantity. ***Work-in-process*** (*WIP*) is the quantity of work parts that have been started in the production sequence but are not yet completed. *WIP* consists of both the parts that are currently being processed (e.g., parts that are fixtured in the machines) as well as those that are between processing steps (e.g., parts waiting to be loaded into machines). WIP does not include parts that have been completed, nor does it include the starting raw materials.

In the Toyota Production System, the standard work-in-process quantity is the minimum number of parts necessary to avoid situations where workers are waiting. For example, if the first worker in Figure 7 had only one work part for machines 1 and 2, he would have to wait for machine 1 to finish its cycle before moving the part to machine 2. But if there were two work parts, one in front of each machine, then he could load the first machine and move immediately to load the second machine without waiting for the first machine to finish.

[10]Based on an example in [12].

Standard Operations Routine Sheet					Worker 1				Cycle Time = 1.6 min			
Operation	Operation Time											
1												
2												
9												
Time (min) →	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4

Standard Operations Routine Sheet					Worker 2				Cycle Time = 1.6 min			
Operation	Operation Time											
3												
7												
8												
Time (min) →	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4

Standard Operations Routine Sheet					Worker 3				Cycle Time = 1.6 min			
Operation	Operation Time											
4												
5												
6												
Time (min) →	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4

Figure 8 The standard operations routine sheets for the three workers in Figure 7.

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REVIEW QUESTIONS

1 Define lean production.
2 Name the two pillars of the Toyota Production System.
3 What is the Japanese word for waste?
4 Name the seven forms of waste in production, as identified by Taiichi Ohno.
5 What is meant by the term *just-in-time production system*?
6 What is the objective of a just-in-time production system?
7 What is the difference between a push system and a pull system in production control?
8 What is a kanban? What are the two types of kanban?
9 What is the basic starting point in a study to reduce setup time?
10 What does production leveling mean?
11 How is production leveling accomplished?
12 What does autonomation mean?
13 What does total productive maintenance mean?
14 What is availability as a reliability measure and how is it determined?
15 What does the Japanese word "kaizen" mean?
16 What is a quality circle?
17 What is visual management?
18 What is an andon board?
19 What is the 5S system?
20 What is takt time?
21 What are standardized work procedures in the Toyota Production System?

PROBLEMS

Setup Time Reduction

1 A stamping plant supplies sheet metal parts to a final assembly plant in the automotive industry. The following data are values representative of the parts made at the plant. Annual demand is 150,000 pc (for each part produced). Average cost per piece is $18 and holding cost is 20% of piece cost. Changeover (setup) time for the presses is 5 hr and the cost of downtime on any given press is $200/hr. Determine (a) the economic batch size and (b) the total annual inventory cost for the data. If the changeover time for the presses could be reduced to 10 min, determine (c) the economic batch size and (d) the total annual inventory cost.

2 A supplier of parts to an assembly plant in the household appliance industry is required to make deliveries on a just-in-time basis (daily). For one of the parts that must be delivered, the daily requirement is 200 parts, 5 days/week, 52 weeks/year. However, the supplier cannot afford to make just 200 parts each day; it must produce in larger batch sizes and maintain an inventory of parts from which 200 units are withdrawn for shipment each day. Cost per piece is $20 and holding cost is 24% of piece cost. Changeover time for the production machine used to produce the part is 2 hr and the cost of downtime on this machine is $250/hr. Determine (a) the economic batch size and (b) the total annual inventory cost for the data. The supplier hopes to reduce the batch size from the value determined in part (a) to 200 units, consistent with the daily quantity delivered to the appliance assembly plant. (c) Determine the changeover time that would allow the economic batch size in stamping to be 200 pieces. (d) What is the corresponding total annual inventory cost for this batch size, assuming the changeover time in part (c) can be realized?

3 Monthly usage rate for a certain part is 15,000 units. The part is produced in batches and its manufacturing costs are estimated to be $7.40. Holding cost is 20% of piece cost. Currently the production equipment used to produce this part is also used to produce 19 other parts with similar usage and cost data (assume the data to be identical for purposes of this problem). Changeover time between batches of the different parts is now 4.0 hr, and cost of downtime on the equipment is $275/hr. A proposal has been submitted to fabricate a fast-acting slide mechanism that will permit the changeovers to be completed in just 5.0 min. Cost to fabricate and install the slide mechanism is $150,000. (a) Is this cost justified by the savings in total annual inventory cost that would be achieved by reducing the economic batch quantity from its current value based on a 4-hour setup to the new value based on a 5-minute setup? (b) How many months of savings are required to pay off the $150,000 investment?

4 An injection-molding machine is used to produce 25 different plastic molded parts in a typical year. Annual demand for a typical part is 20,000 units. Each part is made out of a different plastic (the differences are in type of plastic and color). Because of the differences, changeover time between parts is significant, averaging 5 hr to (1) change molds and (2) purge the previous plastic from the injection barrel. One setup person normally does these two activities sequentially. A proposal has been made to separate the tasks and use two setup persons working simultaneously. In that case, the mold can be changed in 1.5 hr and purging takes 3.5 hr. Thus, the total downtime per changeover will be reduced to 3.5 hr from the previous 5 hr. Downtime on the injection-molding machine is $200/hr. Labor cost for setup time is $20/hr. Average cost of a plastic molded part is $2.50, and holding cost is 24% annually. For the 5-hour setup, determine (a) the economic batch quantity, (b) the total number of hours per year that the injection-molding machine is down for changeovers, and (c) the annual inventory cost. For the 3.5-hour setup, determine (d) the economic batch quantity, (e) the total number of hours per year that the injection-molding machine is down for changeovers, and (f) the annual inventory cost.

5 Solve the previous problem, but assume that a second proposal has been made to reduce the purging time of 3.5 hr during a changeover to less than 1.5 hr by sequencing the batches of parts so as to reduce the differences in plastic type and color between one part and the next. In the ideal, the same plastic can be used for all parts, thus eliminating the necessity to purge the injection barrel between batches. Thus, the limiting task in changing over the machine is the mold change time, which is 1.5 hr. For the 1.5-hour setup, determine (a) the economic batch quantity, (b) the total number of hours per year that the injection-molding machine is down for changeovers, and (c) the annual inventory cost.

Availability, Utilization, and Fraction Defect Rate

6 Data have been collected on the malfunctions and breakdowns that have caused line stoppages on a certain automated production line. Based on these data, the mean time between failures is 4.37 hr, and the mean time to repair is 24.1 min. Determine the availability of this line.

7 An automated guided vehicle system (AGVS) has been installed recently in a production warehouse. The company that supplied the AGVS claims that its system is very reliable and has an availability of 95%. In the first week of operation, the mean time between malfunctions that caused downtime averaged 12.3 hr/vehicle. On average, how long was each vehicle out of service if the supplier's claim of 95% availability is correct?

8 Three automatic screw machines are dedicated to the production of a certain high-demand fastener. The machines operate 40 hr/week and each machine produces at a rate of 73.2 parts/hr when producing. (a) How many parts can the three machines produce each week at full capacity? (b) If, during a particular week of interest, the three machines actually produced 7514 parts, how many hours of downtime did the three machines experience? (c) In part (b), if 9.25 hr are known to have been due to equipment malfunctions, what was the utilization of the three machines (due to reasons other than reliability problems)?

9 A certain production machine has an availability of 97%, and its utilization is 92%. The fraction defect rate of the parts made on the machine is 0.025, and it operates at only 85% of its rated speed. What is the overall equipment effectiveness of this machine?

Takt Time and Cycle Time

10 The weekly demand for a certain part is 950 units. The plant operates 5 days/week, with an effective operating time of 440 min/day. Determine the takt time for this part.

11 The monthly usage for a component supplied to an appliance assembly plant is 5000 parts. There are 21 working days in the month and the effective operating time of the plant is 450 min/day. Currently, the defect rate for the component is 2.2%, and the equipment used to produce the part is down for repairs an average of 22 min/day. Determine the takt time for this part.

12 The monthly demand for a part produced for an automotive final assembly plant is 16,000 units. There are 20 working days in February and the effective operating time of the plant is 900 min/day (two shifts). The fraction defect rate for the component is 0.017, and the automated machine that produces the part has an availability of 96%. Determine the takt time for this part.